PHYS 209: Aspects of Modern Physics

Module 4: Qubits

Lecture 5: Quantum Algorithms

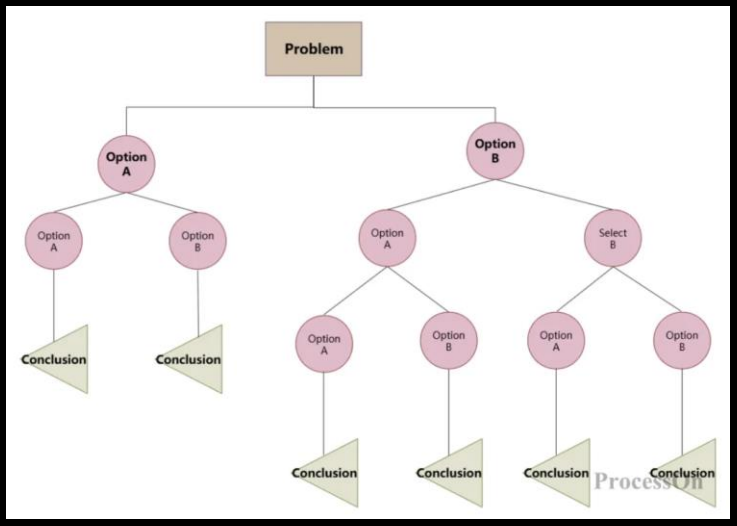
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Summary of Module

- Qubits Math: Linear Algebra, Braket Notation
- Quantum Sensing: Ramsey Sequence
- Quantum Foundations: Entanglement, Bell’s Inequalities
- Quantum Communication: Key Distribution, Teleportation
- Quantum Algorithms

What is an Algorithm?

- Algorithm: Step-by-step procedure for solving a problem.
- Algorithm Complexity: Number of resources (for example, number of steps) required.
- Algorithms can be represented as Decision Trees.
- Query Complexity: Depth of Decision Tree = maximum number of Queries N.
- Goal of Algorithm Design: Minimize query complexity.



Searching Algorithms

- Problem: Find a specific name in a list of N entries.
- Algorithm 1 - Linear Search: check entries one by one until found.
  - Decision tree: chain of depth N.
  - Query complexity: $\mathcal{O}(N)$ - scales linearly with list size
- Algorithm 2 – Binary Search (for sorted lists): Check middle entry, discard half, repeat.
  - Decision tree: binary tree of depth $\log_2 N$
  - Query complexity: $\mathcal{O}(\log_2 N)$ - far fewer queries for large N
- $\mathcal{O}(\cdot)$ captures complexity scaling with N
  - Constants are ignored, dominant scaling kept.
  - $N = 10^6$
    - Linear search complexity $\mathcal{O}(N) = 10^6$
    - Binary search has complexity $\mathcal{O}(\log N) = \log_2 10^6 \approx 20$

The Oracle

- In both algorithms, the list is never fully visible.
  - We access one entry at a time.
  - That is, we query for entries based on index number.
  - In computation, an oracle is a black box that answers queries about the input.

oracle

Query: What is entry i? → Answer: $A[i]$

- Query model of complexity:

$$\text{Algorithm complexity} = \text{number of oracle calls}$$

- Can a quantum algorithm solve a problem using fewer oracle calls than any classical algorithm – even in principle?

How can Quantum Mechanics help?

- Classical Oracle: Takes a single input and returns a single output:

$$\text{Query } x = 0 \rightarrow f(0), \qquad \text{Query } x = 1 \rightarrow f(1)$$

  - Two separate queries required.
- Quantum Oracle: Can act on superposition of inputs:

$$\text{Query } \frac{|0\rangle + |1\rangle}{\sqrt{2}} \rightarrow \text{information about both } f(0) \text{ and } f(1)$$

  - One query required.
  - Quantum parallelism – both values probed in a single oracle call.
- The catch: Measurement collapses superposition.
  - Either get $f(0)$ or $f(1)$.
  - No better than classical?

Deutsch Problem

- Simplest possible Oracle: Function in single input bit
$$f: \{0,1\} \rightarrow \{0,1\}$$
- How many such functions exist?
- All possibilities:
  - Constant: $f(0) = f(1)$
  - Balanced: $f(0) \neq f(1)$
- General problem: Can you determine which function corresponds to the oracle?
  - Requires 2 queries – first ask $f(0)$, then ask $f(1)$.
- Deutsch problem: Can you determine whether the function implemented by the oracle is constant or balanced?
  - Classical algorithm: Still 2 queries. → What about a quantum algorithm?

f	$f(0)$	$f(1)$	B or C
f_1	0	0	Constant
f_2	1	1	Constant
f_3	0	1	Balanced
f_4	1	0	Balanced

Quantum Approach to Deutsch Problem

- Deutsch Problem: Is $f: \{0,1\} \rightarrow \{0,1\}$ constant or balanced?
- Can we query a quantum superposition?

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

- Problem: Constant function f is irreversible.
  - Suppose there is a unitary operator $\hat{U}_{f_1}$ that implements f_1:

$$\hat{U}_{f_1}|0\rangle = |0\rangle, \qquad \hat{U}_{f_1}|1\rangle = |0\rangle$$

  - Since $\hat{U}_f$ is unitary, $\hat{U}_{f_1}^\dagger \hat{U}_{f_1} = \hat{I}$, the above equations give

$$\hat{U}_{f_1}^\dagger \hat{U}_{f_1}|0\rangle = \hat{U}_f^\dagger|0\rangle, \qquad \hat{U}_{f_1}^\dagger \hat{U}_{f_1}|1\rangle = \hat{U}_{f_1}^\dagger|0\rangle$$

$$\hat{I}|0\rangle = \hat{U}_{f_1}^\dagger|0\rangle, \qquad \hat{I}|1\rangle = \hat{U}_{f_1}^\dagger|0\rangle$$

  - Since $\hat{U}_{f_1}^\dagger|0\rangle$ cannot both be $|0\rangle$ and $|1\rangle$, we must conclude that a unitary operator $\hat{U}_{f_1}$ implementing f_1 cannot exist.
  - Same applies for f_2.
- What is needed: Unitary implementation of constant f.

f	$f(0)$	$f(1)$	B or C
f_1	0	0	Constant
f_2	1	1	Constant
f_3	0	1	Balanced
f_4	1	0	Balanced

Unitary Implementation of Oracle

- Question: How to implement constant f using a unitary operator?
- Solution: Use ancilla to store the output and implement as a two-qubit operator.

$$\hat{U}_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$$

  - $\oplus$ is addition mod 2 = XOR gate
  - This preserves input $|x\rangle$.
    - Hence reversibility is guaranteed:

$$y \oplus f(x) \oplus f(x) = y$$

$$\hat{U}_f\hat{U}_f|x\rangle|y\rangle = \hat{U}_f|x\rangle|y \oplus f(x)\rangle = |x\rangle|y \oplus f(x) \oplus f(x)\rangle = |x\rangle|y\rangle$$

    - So, $\hat{U}_f^\dagger = \hat{U}_f$ and it implements the function of oracle on input $|x\rangle$ and outputs it onto $|y\rangle$.

Applying $\hat{U}_f$ twice

y	$f(x)$	$y \oplus f(x)$	$y \oplus f(x) \oplus f(x)$
0	0	$0 \oplus 0 = 0$	$0 \oplus 0 = 0$
0	1	$0 \oplus 1 = 1$	$1 \oplus 1 = 0$
1	0	$1 \oplus 0 = 1$	$1 \oplus 0 = 1$
1	1	$1 \oplus 1 = 0$	$0 \oplus 1 = 1$

y	$f(x)$	$y \oplus f(x)$
0	$f(x)$	$0 \oplus f(x) = f(x)$
1	$f(x)$	$1 \oplus f(x) = 1$

- So, we have a unitary implementation of the Oracle.
  - Question 1: Which state to use as ancilla $|y\rangle$?
  - Question 2: Does using superposition for $|x\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ solve Deutsch problem with just 1 query?

Querying Oracle with Superposition State

- We have the unitary oracle now: $\widehat{U}_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$.
- How many oracle calls needed with query qubit $|x\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$?
- Try ancilla $|y\rangle = |0\rangle$:

$$\widehat{U}_f|x\rangle|y\rangle = \widehat{U}_f \frac{|0\rangle + |1\rangle}{\sqrt{2}}|0\rangle = \frac{|0\rangle|0 \oplus f(0)\rangle + |1\rangle|0 \oplus f(1)\rangle}{\sqrt{2}}$$

$$= \frac{|0\rangle|f(0)\rangle + |1\rangle|f(1)\rangle}{\sqrt{2}}$$

  - Both $f(0)$ and $f(1)$ are encoded in one query – quantum parallelism.
  - But measurement collapses state – we learn one value at a time.
- Question: Can we use a superposition ancilla?

Querying with Superposition Ancilla

- Try ancilla $|y\rangle = |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$:

$$\widehat{U}_f|x\rangle|y\rangle = \widehat{U}_f|x\rangle\frac{|0\rangle - |1\rangle}{\sqrt{2}} = \frac{|x\rangle|f(x)\rangle - |x\rangle|1 \oplus f(x)\rangle}{\sqrt{2}}$$

$$= |x\rangle\frac{|f(x)\rangle - |1 \oplus f(x)\rangle}{\sqrt{2}}$$

- Two possibilities:
  - $f(x) = 0$: $\widehat{U}_f|x\rangle|y\rangle = |x\rangle\frac{|0\rangle - |1\oplus 0\rangle}{\sqrt{2}} = |x\rangle\frac{|0\rangle - |1\rangle}{\sqrt{2}} = |x\rangle|-\rangle$
  - $f(x) = 1$: $\widehat{U}_f|x\rangle|y\rangle = |x\rangle\frac{|1\rangle - |1\oplus 1\rangle}{\sqrt{2}} = |x\rangle\frac{|1\rangle - |0\rangle}{\sqrt{2}} = -|x\rangle|-\rangle$
- General: $\widehat{U}_f|x\rangle|y\rangle = (-1)^{f(x)}|x\rangle|-\rangle$
  - Ancilla is unchanged.
  - Value of $f(x)$ in encoded in phase.

Superposition State + Superposition Ancilla

- Ancilla $|y\rangle = |-\rangle$:
- $|x\rangle = |0\rangle$ or $|1\rangle$: $\widehat{U}_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$
  - Ancilla unchanged.
  - Value of $f(x)$ encoded in phase.
- $|x\rangle = |+\rangle$: $\widehat{U}_f|+\rangle|y\rangle = \frac{(-1)^{f(0)}|0\rangle+(-1)^{f(1)}|1\rangle}{\sqrt{2}}|-\rangle$
  - f constant: $(-1)^{f(0)} = (-1)^{f(1)}$ - same phase on both terms.
  - f balanced: $(-1)^{f(0)} \neq (-1)^{f(1)}$ - opposite phase on both terms.
- The global property of constancy/balance is in the phase difference.

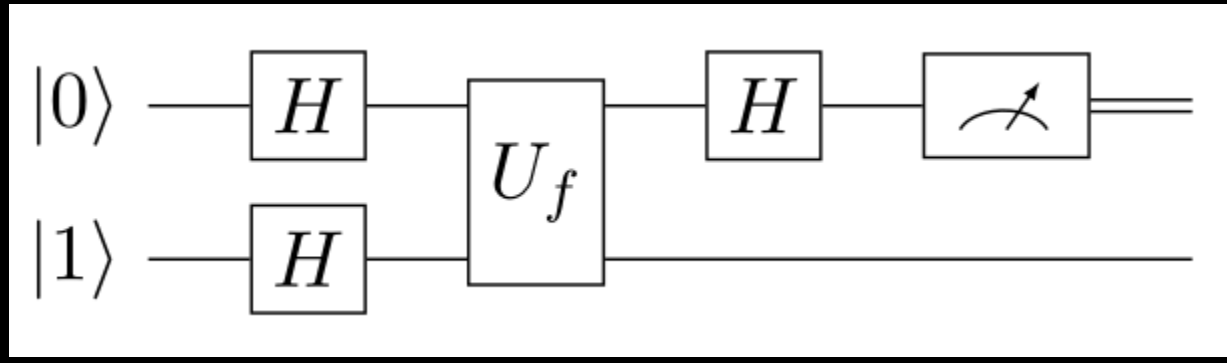
Measurement

- After oracle query:

$$\widehat{U}_f|+\rangle|y\rangle = \frac{(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle}{\sqrt{2}}|-\rangle$$

  - Constant f: $\widehat{U}_f|+\rangle|y\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}|-\rangle = |+\rangle|-\rangle$
  - Balanced f: $\widehat{U}_f|+\rangle|y\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}|-\rangle = |-\rangle|-\rangle$
- Rotate state by $\frac{\pi}{2}$ - use Hadamard gate $\widehat{H} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$
  - $\widehat{H}|+\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\ 1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}2\\ 0\end{bmatrix} = \begin{bmatrix}1\\ 0\end{bmatrix} = |0\rangle$
  - $\widehat{H}|-\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\ -1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}0\\ 2\end{bmatrix} = \begin{bmatrix}0\\ 1\end{bmatrix} = |1\rangle$
- Measure state along z:
  - $|0\rangle$: Constant f
  - $|1\rangle$: Balanced f

Deutsch Algorithm



https://prefetch.eu/know/concept/deutsch-jozsa-algorithm/

- Initial State:

$$|\psi_1\rangle = |0\rangle|1\rangle$$

- Hadamard Gate:

$$|\psi_2\rangle = \left(\hat{H} \otimes \hat{H}\right)|\psi_1\rangle = |+\rangle|-\rangle$$

- Oracle Query:

$$|\psi_3\rangle = \hat{U}_f|\psi_2\rangle = \frac{(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle}{\sqrt{2}}|-\rangle$$

- Hadamard Gate:

$$|\psi_4\rangle = \left(\hat{H} \otimes \hat{I}\right)|\psi_3\rangle = \begin{cases} |0\rangle|-\rangle & \text{Constant} \\ |1\rangle|-\rangle & \text{Balanced} \end{cases}$$

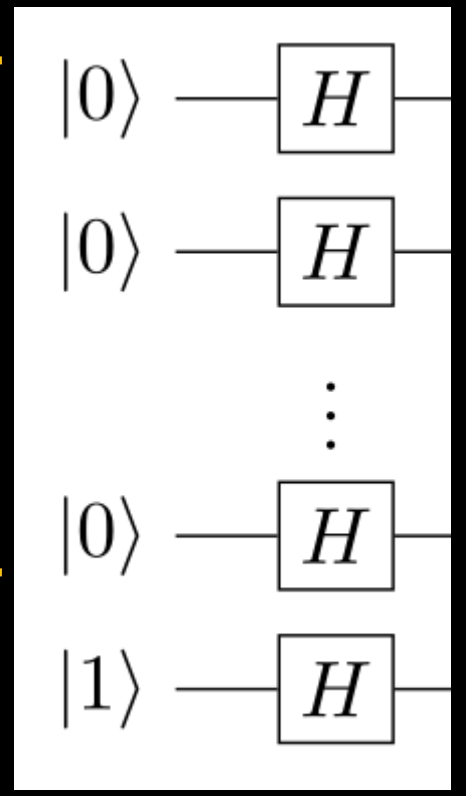
- Measurement:

$$|x\rangle = |0\rangle: \quad \text{Constant}$$
$$|x\rangle = |1\rangle: \quad \text{Balanced}$$

Deutsch Algorithm - Summary

- Deutsch Problem: Tell whether $f:\{0,1\} \rightarrow \{0,1\}$ is constant $f(0) = f(1)$ or balanced $f(0) \neq f(1)$.
- Deutsch Problem can be solved using
  - Classical Algorithm requiring two queries
  - Quantum Algorithm requiring one query
- Relies on $\frac{\pi}{2}$ pulse $\rightarrow$ interference $\rightarrow$ $\frac{\pi}{2}$ pulse $\rightarrow$ measurement
  - Analogous to Ramsey sequence.

Deutsch-Jozsa Problem



- Generalize to n-bit input: $f: \{0,1\}^n \rightarrow \{0,1\}$
  - Number of possible input: 2^n
- Assume n is even and that f is either:
  - Constant: same output for all 2^n inputs
  - Balanced: half outputs give 0, half outputs give 1
    - Half outputs: $2^n/2 = 2^{n-1}$
- Deutsch-Jozsa Problem: Determine whether f is constant or balanced.
- Classical query complexity:
  - Worst case: must check $2^{n-1} + 1$ inputs
  - Query complexity: $\mathcal{O}(2^n)$ - exponential in n
- What about with a quantum algorithm?

Deutsch vs. Deutsch-Jozsa

- Initial State:

$$|\psi_1\rangle = |0\rangle|1\rangle$$

- Hadamard Gate:

$$|\psi_2\rangle = \left(\hat{H} \otimes \hat{H}\right)|\psi_1\rangle = |+\rangle|-\rangle$$

- Oracle Query:

$$|\psi_3\rangle = \hat{U}_f|\psi_2\rangle = \frac{(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle}{\sqrt{2}}|-\rangle$$

- Hadamard Gate:

$$|\psi_4\rangle = \left(\hat{H} \otimes \hat{I}\right)|\psi_3\rangle = \begin{cases} |0\rangle|-\rangle & \text{Constant} \\ |1\rangle|-\rangle & \text{Balanced} \end{cases}$$

- Measurement:

$|x\rangle = |0\rangle$: Constant

$|x\rangle = |1\rangle$: Balanced

- Initial State:

$$|\psi_1\rangle = |0\rangle^{\otimes n}|1\rangle$$

- Hadamard Gates on input qubits:

$$|\psi_2\rangle = \left(\hat{H}^{\otimes n} \otimes \hat{H}\right)|\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle|-\rangle$$

- Oracle Query:

$$|\psi_3\rangle = \hat{U}_f|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} (-1)^{f(x)}|x\rangle|-\rangle$$

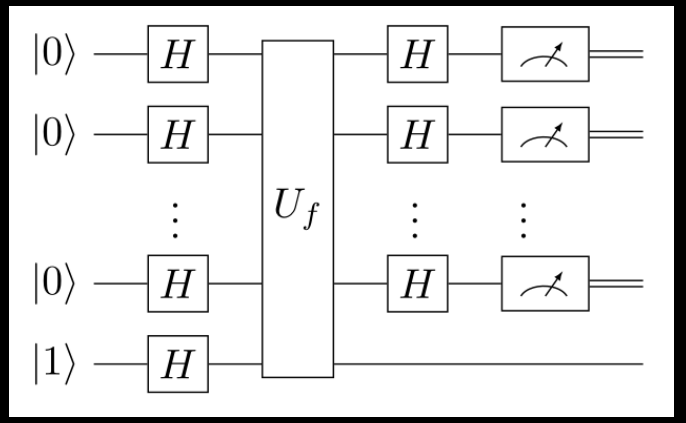
- Hadamard Gates on input qubits:

$$|\psi_4\rangle = \left(\hat{H}^{\otimes n} \otimes \hat{I}\right)|\psi_3\rangle$$
$$= \sum_{z\in\{0,1\}^n} \left(\frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{f(x)+x\cdot z}\right)|z\rangle|-\rangle$$

- Measurement:

$$P\left(|0\rangle^{\otimes n}\right) = \left|\frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{f(x)}\right|^2 = \begin{cases} 1 & \text{Constant} \\ 0 & \text{Balanced} \end{cases}$$

Deutsch-Jozsa Algorithm - Summary



- Deutsch-Jozsa algorithm determines whether $f: \{0,1\}^n \rightarrow \{0,1\}$ is constant or balanced with 1 oracle query with complete certainty.

  Measure $|0\rangle^{\otimes n} \rightarrow f$ constant

  Measure anything else $\rightarrow f$ balanced

- Query complexity comparison:

Algorithm	Classical (Deterministic)	Classical (Probabilistic)	Quantum
Deutsch	2	2	1
Deutsch-Jozsa	$\mathcal{O}(2^n)$	$\mathcal{O}(1)$	1

- Quantum advantage over deterministic classical algorithm is exponential.

Other Quantum Algorithms

- Grover Algorithm: Search for an item in unsorted list of N items
  - Classical: $\mathcal{O}(N)$ Quantum: $\mathcal{O}\left(\sqrt{N}\right)$
- Quantum Simulation
  - Classical: Exponentially difficult Quantum: Natural platform
- Shor's Algorithm: Factorize integer
  - Classical: No known polynomial algorithm Quantum: $\mathcal{O}\left((\log N)^3\right)$
  - Breaks RSA Encryption
    - With ~ 20,000 perfect qubits (logical)
    - With ~ millions of noisy qubits (physical)
  - Current state of the art:
    - $\mathcal{O}(10^3)$ physical qubits
    - $\mathcal{O}(10^1)$ logical qubits

Outlook

- Qubits use quantum superposition as a resource.

	Physical System	Superposition	Phase Imprint	Interference
Sensing	Spin Qubit	$\pi/2$ pulse	Free Precession	$\pi/2$ pulse
Algorithms	Query Register	$\hat{H}^{\otimes n}$	Oracle $\hat{U}_f$	$\hat{H}^{\otimes n}$

- Quantum parallelism → quantum advantage.
- Challenge: Prone to noise / errors.
  - Need to shield qubit from noise/error.
  - Need to correct for noise/error.